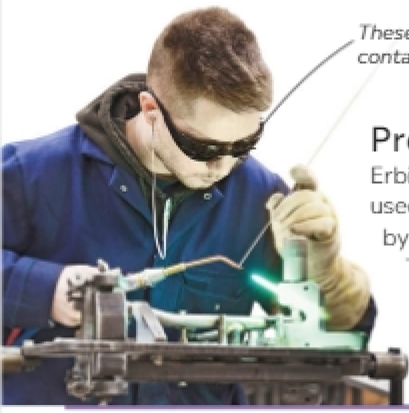
Erbium

68
Er

This silvery metal is used in the form of its compound erbium oxide to add a pink color to glassware, ceramic glazes, and cubic zirconia gemstones. It's also used to make lens filters for photography and for certain types of laser.

Laboratory sample of pure erbium



These safety goggles contain erbium.

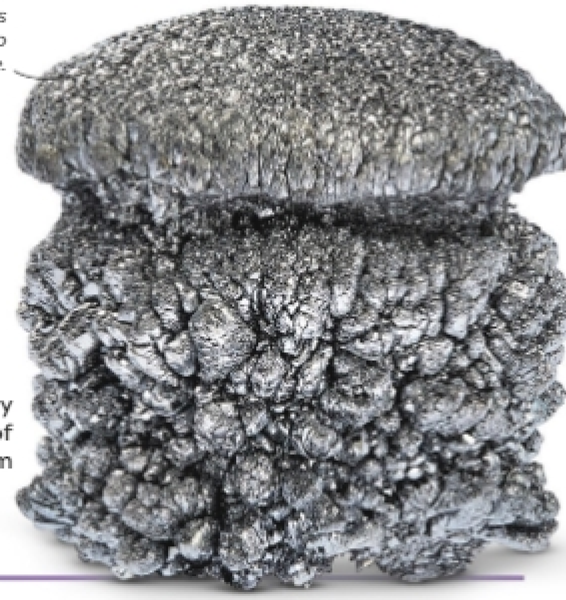
Protective goggles

Erbium oxide is added to glass used in protective goggles worn by welders and glassblowers. This glass absorbs the ultraviolet radiation that can damage our eyes.

Thulium

69
Tm

Not to be confused with the similarly named thallium, pure thulium was first isolated in 1911 by the Swedish chemist Per Teodor Cleve. It's found in X-ray machines used by doctors and dentists.



This metal is soft enough to cut with a knife.

Laboratory sample of pure thulium

Ytterbium

70
Yb

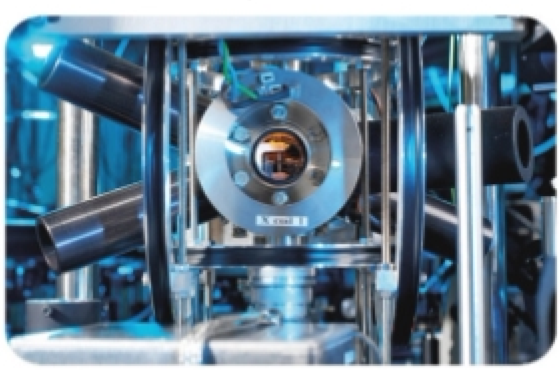
This soft metal reacts with air to form a layer of golden-brown oxide compound on its surface. A small amount of ytterbium is sometimes used to strengthen steel. Its compounds are also used in lasers.

Laboratory sample of ytterbium

Ytterbium develops a golden color in air.

Standardizing time

Inside this highly sensitive optical clock, ytterbium atoms oscillate in response to the light of a laser beam. Counting these oscillations helps keep time accurately.



Optical clock, National Physical Laboratory, Teddington, UK

Lutetium

71
Lu

Like most other lanthanides, lutetium's principal ore is also monazite. This element is a hard, silver-colored metal with a high density. There's more lutetium than silver in Earth's crust, but lutetium is very difficult to separate from other elements, so it's not widely used. Small amounts are used by the petroleum industry, and one of its radioactive isotopes—lutetium-177—is used in cancer therapy.